Individual Case Safety Report Assessment

Simvastatin and clarithromycin associated rhabdomyolysis

Executive assessment

This is a serious suspected adverse drug reaction associated with the concurrent use of simvastatin and clarithromycin. The patient developed marked muscle injury with myoglobinuria, acute kidney injury, and hyperkalemia, requiring hospital admission and intravenous fluids. The temporal relationship, objective findings, improvement after withdrawal, and established CYP3A4 interaction support causality. The combination-level WHO–UMC assessment is Probable/Likely; the Naranjo score is 7 (Probable), with the caveat that the scale is not designed to fully resolve interacting-drug causality.

ICSR assessment item	Conclusion
Reaction	Rhabdomyolysis with myalgia, proximal muscle weakness, myoglobinuria, acute kidney injury, and hyperkalemia.
Seriousness	Serious: inpatient hospitalization; medically important renal and electrolyte complications.
Clinical severity	Severe acute reaction based on marked CK elevation, functional weakness, renal injury, and need for inpatient treatment.
Expectedness	Expected / known interaction risk in the referenced product information; formal listedness must be checked against the applicable local label version.
Outcome	Recovered / resolved by Day 14 per source narrative.
Reporting disposition	Potentially reportable serious suspected ICSR. The supplied excerpt lacks primary reporter identity/qualification and country, so follow-up is needed before it is submission-ready under ICH minimum criteria. Record and process per local SOP; do not delay safety evaluation while obtaining missing data.

1. Case narrative

A 68-year-old woman weighing 58 kg had hypertension and dyslipidemia. Baseline serum creatinine was 0.9 mg/dL, and she had no previous muscle-related complaints. She had taken simvastatin 40 mg at bedtime and amlodipine 5 mg once daily for three years, and hydrochlorothiazide 12.5 mg once daily for two years. Clarithromycin 500 mg twice daily was newly prescribed for seven days for a lower respiratory tract infection. The prescriber was unaware that she was taking simvastatin.

Day	Event / action
0	Clarithromycin started.
3	Mild thigh and calf pain developed; initially attributed to the infection.
4	Marked proximal muscle weakness, difficulty climbing stairs, and cola-coloured urine.
6	Hospital admission; clarithromycin and simvastatin discontinued.
6–10	Intravenous fluids and electrolyte monitoring.
10	CK began trending down and urine colour cleared.
14	Creatinine improved to 1.0 mg/dL and muscle weakness resolved.

2. Clinical evidence and reaction identification

Finding at admission (Day 6)	Result	Reference range	Safety interpretation
Creatine kinase (CK)	18,500 U/L	<170 U/L	Marked skeletal muscle injury; supports rhabdomyolysis.
Serum creatinine	1.9 mg/dL	0.5–1.1 mg/dL	Acute kidney injury relative to baseline 0.9 mg/dL.
Potassium	5.6 mmol/L	3.5–5.0 mmol/L	Hyperkalemia, consistent with muscle-cell injury.
AST / ALT	420 / 210 U/L	<40 / <40 U/L	Can rise with skeletal muscle injury; alone does not establish primary liver injury.
Urine myoglobin	Positive	Negative	Supports myoglobin release.
TSH	Normal	Within normal range	Makes hypothyroidism less likely.

The source states that trauma, strenuous exercise, alcohol use, and seizures were excluded. The very high CK, positive urine myoglobin, muscle symptoms, and dark urine are consistent with rhabdomyolysis. The elevated creatinine and potassium indicate clinically important complications. The infection may explain nonspecific aches but does not adequately account for this full pattern.

Suspected and concomitant medicines

Medicine	Dose and exposure	Role in assessment
Clarithromycin	500 mg twice daily; newly initiated Day 0 for 7 days	Precipitating interacting medicine; strong CYP3A4 inhibition can increase simvastatin exposure.
Simvastatin	40 mg at bedtime; used for 3 years	Affected / object medicine; CYP3A4 substrate associated with myotoxicity at increased exposure.
Amlodipine	5 mg once daily; used for 3 years	Contributing risk factor noted in source: simvastatin 40 mg/day exceeded the source-stated 20 mg/day maximum with amlodipine.
Hydrochlorothiazide	12.5 mg once daily; used for 2 years	Concomitant medicine; no clear case-specific evidence identifies it as a primary suspect.

3. ICSR data review and case validity

The report contains the clinical event, suspected/interacting medicines, and an identifiable patient description (age and sex).

Core / follow-up data element	Status in source	DSA action
Patient	68-year-old woman; weight 58 kg.	Capture available patient characteristics; seek permitted patient identifier/date of birth and relevant medical history.
Primary source reporter	Prescriber is described, but not identified as the reporter.	Obtain reporter identity/initials or other verification, qualification, and country; document direct knowledge/source.

Suspect/interacting products	Simvastatin and clarithromycin identified; doses and durations supplied.	Confirm brand/manufacturer, formulation, route, start/stop dates, indication, and whether interaction was suspected by reporter.
Reaction/event	Rhabdomyolysis pattern and clinical features supplied.	Record verbatim terms and coded reaction concepts using current MedDRA conventions; do not infer diagnosis beyond source support.
Seriousness/outcome	Hospital admission and complications supplied; recovered by Day 14.	Record hospitalization criterion, medically important complications, treatment, and outcome. Clarify discharge date and sequelae.
Case administration	Report receipt date, country of receipt, case number, and duplicate status not supplied.	Establish source, receipt date/day zero, country, duplicate check, and applicable reporting pathway.
Medical history / risk factors	Hypertension, dyslipidemia, baseline creatinine, no prior muscle symptoms; TSH normal.	Follow up renal history, hydration status, infection details, alcohol/exercise, allergies, and other relevant risk factors if available.
Product information version	Not specified.	Check current local reference safety information for expectedness/listedness for each suspect product and interaction.

4. Causality assessment

Naranjo ADR Probability Scale

Scored for the combined simvastatin–clarithromycin exposure as a suspected interaction-associated reaction. The scale cannot cleanly apportion causality between the two interacting medicines.

Naranjo question	Answer	Score	Case rationale
Previous conclusive reports?	Yes	+1	The interaction and associated myotoxicity are recognized in product safety information.
Event appeared after suspected drug(s) were administered?	Yes	+2	Symptoms began three days after clarithromycin was added to chronic simvastatin.
Improved after discontinuation or specific antagonist?	Yes	+1	Both suspect medicines were stopped and the patient improved with supportive care.
Reaction reappeared on readministration?	Unknown	0	No rechallenge; not required and not reported.
Alternative causes could on their own have caused the reaction?	No	+2	Source reports key alternatives excluded; infection alone does not explain the objective severe muscle injury.
Reaction reappeared with placebo?	Unknown	0	No placebo challenge.
Toxic drug concentration detected?	Unknown	0	No simvastatin concentration was measured/reported.
Dose response demonstrated?	Unknown	0	No dose increase/decrease observation.
Similar reaction on previous	No	0	No previous muscle-related complaints are

exposure?			reported.
Objective evidence confirms the event?	Yes	+1	CK 18,500 U/L, positive urine myoglobin, weakness, renal and electrolyte abnormalities.
Total		7	Probable (5–8).

Interpretation: the score supports a probable drug-related event. The result is an aid to judgment and does not prove single-drug causality. Because both suspect medicines were withdrawn together, separate Naranjo scores would not reliably distinguish the interaction components.

WHO–UMC causality classification

Combined interaction: Probable / Likely. The event has a reasonable temporal relationship to the addition of clarithromycin; a pharmacologically plausible and recognized interaction exists; major alternative causes were investigated; and improvement after withdrawal is clinically reasonable. Rechallenge is not required for this category. For simvastatin or clarithromycin considered separately, Possible is more cautious because both were stopped together and their individual contributions cannot be separated from this case alone.

5. Seriousness, severity, and expectedness

Dimension	Classification	Rationale
Regulatory seriousness	Serious	Required inpatient hospitalization (ICH/EMA seriousness criterion). Acute kidney injury and hyperkalemia are medically important complications. No death or persistent disability is reported.
Clinical severity / intensity	Severe	Marked CK elevation (18,500 U/L), proximal weakness impairing stair climbing, myoglobinuria, acute kidney injury, hyperkalemia, IV fluids, and inpatient monitoring.
Expectedness / listedness	Expected / known interaction risk	Current clarithromycin labeling identifies concomitant simvastatin as contraindicated and describes increased simvastatin concentrations with risk of myopathy including rhabdomyolysis. Confirm the exact local reference label/version for formal listedness.
Outcome	Recovered / resolved	CK declined, urine colour cleared, creatinine improved to 1.0 mg/dL, and weakness resolved by Day 14.

Seriousness and severity describe different things: seriousness is based on regulatory outcomes and medical importance; severity describes intensity. Here the event meets serious criteria because of hospitalization, and it was clinically severe. Expectedness does not reduce seriousness or negate reportability.

6. Reporting recommendation and DSA disposition

- Create/retain the case record and request follow-up promptly to complete the missing reporter qualification, reporter country, source and receipt dates, and product details. Perform duplicate search when case identifiers/source become available.

- Treat as a suspected serious ADR involving an interacting medicinal product combination. Causality need not be proven before the case is processed or reported under applicable requirements.
- If the minimum validation elements are confirmed, submit through the applicable local/regional ICSR pathway within required timelines. ICH E2D(R1) generally describes expedited ICSR submission as soon as possible and no later than 15 calendar days when regional/local criteria for expedited reporting are met; exact requirements depend on jurisdiction and report type.
- Record the seriousness basis as inpatient hospitalization and medically important acute kidney injury/hyperkalemia. Record expectedness against the applicable reference product information, not solely the source article or general drug knowledge.
- Send clinically appropriate follow-up questions to the source: CK and renal/electrolyte trends, exact hospitalization/discharge dates, treatment, final diagnosis, sequelae, medication stop dates, other medicines, and outcome confirmation.
- Escalate as a medication safety / interaction issue for review: medication reconciliation failed to identify simvastatin, clarithromycin was co-prescribed, and simvastatin 40 mg/day with amlodipine exceeded the dose limit cited by the source case.

7. Clinical impact and conclusion

The event caused acute, substantial harm: skeletal muscle breakdown with severe biochemical injury, muscle weakness, dark urine, acute kidney injury, and hyperkalemia. It required hospitalization, IV fluids, and monitoring. The patient recovered by Day 14 without reported persistent impairment. The combination is pharmacologically plausible and consistent with a known, preventable drug interaction.

Final assessment: serious, clinically severe, expected interaction-associated rhabdomyolysis; Naranjo 7 (Probable); WHO–UMC Probable/Likely for the combined exposure. Individual causality for either drug alone remains less certain.